so they can’t get into the hole. By contrast, the whirl grabs photons that are trying to travel opposite to the spin and sucks them into the hole, where they slow the spin. Ultimately, when the hole’s spin reaches 0.998 of the maximum, an equilibrium is reached, with spin-down by the captured photons precisely counteracting spin-up by the accreting gas. This equilibrium appears to be somewhat robust. In most astrophysical environments I expect black holes to spin no faster than about 0.998 of the maximum.

However, I can imagine situations—very rare or never in the real universe, but possible nevertheless—where the spin gets much closer to the maximum, even as close as Chris requires to produce the slowing of time on Miller’s planet, a spin one part in 100 trillion less than the maximum spin. Unlikely, but possible.

This is common in movies. To make a great film, a superb filmmaker often pushes things to the extreme. In science fantasy films such as *Harry Potter*, that extreme is far beyond the bounds of the scientifically possible. In science fiction, it’s generally kept in the realm of the possible. That’s the main distinction between science fantasy and science fiction. *Interstellar* is science fiction, not fantasy. Gargantua’s ultrafast spin is scientifically possible.

Gargantua’s Anatomy

Having determined Gargantua’s mass and spin, I used Einstein’s equations to compute its anatomy. As in the previous chapter, here we focus solely on the external anatomy, leaving the interior (especially Gargantua’s singularities) for Chapters 26 and 28.

In the top picture in Figure 6.3, you see the shape of Gargantua’s equatorial plane as viewed from the bulk. This is like Figure 5.5, but